Year 12 Maths - Binomial expansion

Similar questions - Using the expansion

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Expand $(1+x)^5$. **[1]**

(b) By writing $101 = 100+1$ and using your expansion, find the exact value of 101^5. **[4]**

2 The random variable $X \sim \mathrm{B}\left(6, \frac{2}{3}\right)$. The probabilities $\mathrm{P}(X = r)$ are the terms of the expansion of $(a+b)^6$.

(a) State the values of a and b. **[1]**

(b) Show that $\mathrm{P}(X = 4) = \dfrac{80}{243}$. **[2]**

(c) Find $\mathrm{P}(X \geq 5)$. **[3]**

Question 1

(a) $(1+x)^5 = 1+5x+10x^2+10x^3+5x^4+x^5$

(b) $101^5 = (100+1)^5 = 100^5+5(100)^4+10(100)^3+10(100)^2+5(100)+1 = 10\,000\,000\,000+500\,000\,000+10\,000\,000+100\,000+500+1 = 10\,510\,100\,501$

Question 2

(a) $a = \frac{1}{3}$ and $b = \frac{2}{3}$ (or the other way round, with matching powers)

(b) $\mathrm{P}(X=4) = \binom{6}{4}\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^2 = 15\times\frac{16}{81}\times\frac{1}{9} = \frac{240}{729} = \frac{80}{243}$

(c) $\mathrm{P}(X=5) = 6\left(\frac{2}{3}\right)^5\left(\frac{1}{3}\right) = \frac{192}{729}$ and $\mathrm{P}(X=6) = \left(\frac{2}{3}\right)^6 = \frac{64}{729}$. So $\mathrm{P}(X\geq 5) = \frac{256}{729}$